

aws

Search

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United States (N. Virginia)

Account ID: 4733-0171-3056

AWSLabUser-aPYyVrT2rMDZ9XYrqPVhGD/fd15c6c...

Managed Apache Flink

 >

Apache Flink applications

 >

AnomalyDetection

 >

Edit configuration

Configure AnomalyDetection

Apache Flink runtime - new

Info

Apache Flink runtime

Before updating your runtime, we strongly recommend that you read the documentation. [Learn more](#)

☒ Keep current Apache Flink runtime

Apache Flink 1.15

☐ Change Apache Flink runtime

Application code location

Info

Amazon S3 bucket

Choose the S3 bucket that contains your application's JAR file. Your application code is reloaded from the S3 bucket every time you update your application using this page. To update other application settings without reloading the application code, use the AWS CLI.

s3://kinesis-flink-application-1fb8d000

Browse

Create

Format: s3://bucket

Path to S3 object

Specify the path and object name of the JAR file containing your application code.

AnomalyDetection.jar

Access to application resources

☐ Create / update IAM role **kinesis-analytics-AnomalyDetection-us-east-1** with required policies

☒ Choose from IAM roles that Managed Service for Apache Flink can assume

Service role

lab-apache-flink-role-1fb8d000

Create

View [lab-apache-flink-role-1fb8d000](#) in IAM

Snapshots

Info

If turned on, the service automatically creates snapshots of your application's state to provide exactly-once processing semantics.

Snapshots

☐ Turn on snapshots

Turn on creating service-triggered snapshots when the application stops, updates, or scales.

Application restore configuration

Info

Application isn't currently running. To configure application restore configuration, run this application.

Automatic system rollback - new

Info

The service performs a rollback to the previous running version of the application as soon as it detects an error. Managed Service for Apache Flink continues to process data and maintains high availability for you applications. [Learn more](#)

☒ Turn on system rollback

Checkpoint configuration

Info

A checkpoint is an up-to-date backup of a running application that is used to recover immediately from an unexpected application disruption or failover. For details on checkpointing in Apache Flink applications, see [Checkpoints](#) in the Apache Flink documentation.

Configuration type

Indicates whether the application uses default checkpointing behavior or whether it has been customized.

☒ Use default values

☐ Customize

Checkpoint

On

Checkpoint interval update (in milliseconds)

Describes updates to the interval in milliseconds between checkpoint operations.

60000

Minimum pause between checkpoints (in milliseconds)

Describes updates to the minimum time in milliseconds after a checkpoint operation completes that a new checkpoint operation can start

5000

Encryption - new

Info

Choose the data encryption you want to use. Data management requires encryption.

Incompatible Flink version

Encryption is not available in your current version. Upgrade to Flink versions 1.20 or higher to enable encryption.

chrome-extension://mcbpblocgmgnfpjppndjkmgjaogfceg/fsCaptured.html?id=5

1/3

Scaling [Info](#)

[Request limit increase](#)

Managed Service for Apache Flink provisions capacity as Kinesis Processing Units (KPU). To specify how your analytics application scales, configure the parallel running of tasks (Parallelism) and the allocation of resources (Parallelism per KPU) for tasks. One KPU represents 1 vCPU, 4 GiB of memory, and 50 GiB of running application storage.

Parallelism

An analytics application allows you to run one or more tasks. You can split a task into several parallel instances for running. Each parallel instance processes a subset of the task's data. The number of parallel instances of a task is called its parallelism. Adjusting parallelism helps run your tasks more efficiently.

1

Starting value and minimum value for current parallelism. Minimum: 1, maximum: service limit.

Parallelism per KPU

The number of parallel tasks that can be scheduled per KPU for your analytics application. The default is 1 and the maximum is 8. For analytics applications that have high blocking operations (for example, I/O), a higher Parallelism per KPU allows for full utilization of KPU resources.

1

Minimum: 1, maximum: 8.

KPU

Managed Service for Apache Flink provisions the KPUs that are needed to run your application with the following formula:
KPUs for the application = (Parallelism) divided by (Parallelism per KPU)

1

Automatic scaling

If turned on, Managed Service for Apache Flink elastically scales your application's parallelism to accommodate the data throughput of your source and operator.

☒ Turn on automatic scaling

Logging and monitoring [Info](#)

[Amazon CloudWatch pricing](#)

Managed Service for Apache Flink reports metrics to Amazon CloudWatch metrics and can log record delivery errors to Amazon CloudWatch Logs.

Logging with CloudWatch Logs

Managed Service for Apache Flink can log record delivery errors to CloudWatch Logs. If turned on, a log group and corresponding log stream are created on your behalf. Additional per-GB-month charges apply for collecting, storing, and analyzing logs.

☒ Turn on logging (recommended)

The IAM role associated with the analytics application must have CloudWatchFullAccess policy for writing to CloudWatch Logs.

Monitoring log level (Flink-Log4J)

- ☐ Error
Includes error.
- ☐ Warn
Includes error and warn.
- ☒ Info
Includes error, warn, and info.
- ☐ Debug
Includes error, warn, info, and debug.

Log stream

- ☒ Use the default log stream
Managed Service for Apache Flink will create one if it doesn't exist.
- ☐ Use a custom log stream

Monitoring metrics level with CloudWatch

- ☒ Application
Includes job-level metrics such as uptime, downtime, checkpoints, and MillisBehindLatest.
- ☐ Task
Flink Task level metrics such as numRecordsInPerSecond for the application.
- ☐ Operator
Operator-level metrics such as numRecordsInPerSecond for an operator.
- ☐ Parallelism (not recommended)
Operator-level metrics such as numRecordsInPerSecond for an operator.

Networking [Info](#)

Configure your application to access resources inside a VPC.

VPC connectivity

- ☒ Custom VPC configuration
Allows you to configure your application to use any of your VPCs.
- ☐ VPC configuration based on Amazon MSK cluster
Allows you to choose the VPC based on an MSK cluster.

VPC

Choose the VPC in which you want your application to access resources.

No VPC



[Create](#)

Runtime properties [Info](#)

You can also group application properties into multiple groups. These are useful to store configuration settings without the need to change application code.

Group ID

Key

Value

lab

inputStreamName

WindDataStream

[Remove](#)

lab

outputStreamName

AnomalyDetectionStream

[Remove](#)

lab

region

us-east-1

[Remove](#)

[Add new item](#)**Application maintenance** [Info](#)

Security patches for your application's infrastructure are automatically applied during your maintenance window. Your application may face downtime of a few seconds during maintenance. Actual downtime is dependent on the specific application.

Maintenance window

- ☒ 03:00 (Default for United States (N. Virginia) us-east-1)
- ☐ Custom maintenance window

Maintenance window end time

11:00 UT

[Cancel](#)[Save changes](#)